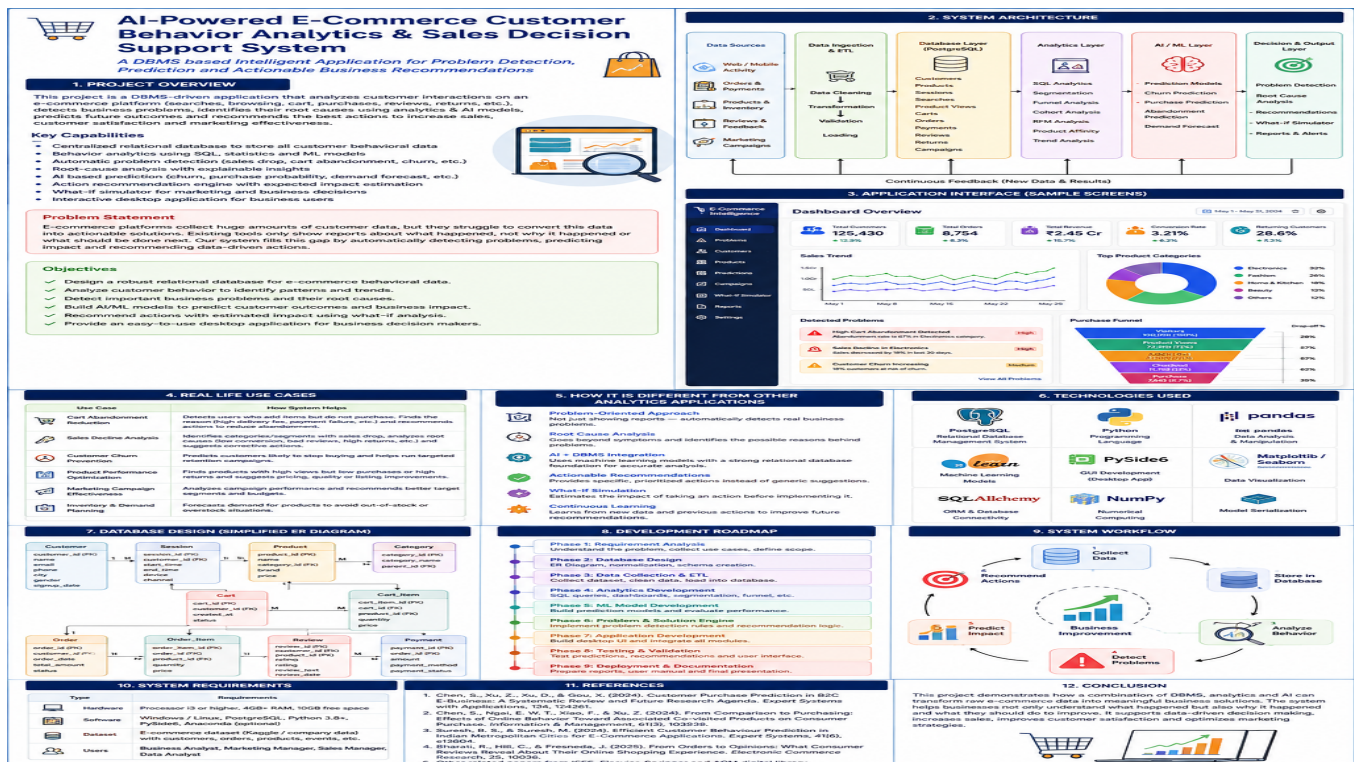


AI-POWERED E-COMMERCE CUSTOMER BEHAVIOR ANALYTICS & SALES DECISION SUPPORT SYSTEM

A DBMS-centered application for customer behavior analysis, problem detection, prediction and actionable business recommendations



Project level: SY Engineering / 3rd Semester DBMS Course

Recommended implementation: Desktop application + PostgreSQL + Python analytics/ML

Purpose: Academic project with a research-paper / patent-exploration pathway

1. Executive Summary

This project is proposed as a DBMS-driven intelligent application rather than a conventional e-commerce website. The system stores customer, product, session, event, cart, order, review and campaign information in a structured relational database. It then converts this data into behavioral analytics, predictions, detected business problems and evidence-based recommendations.

The central design principle is Data → Insight → Prediction → Action. The application should not merely report what happened. It should identify meaningful changes in customer behavior, explain which customer/product segment is affected, estimate likely outcomes, and recommend an intervention that a business manager can investigate or test.

The accompanying research material identified recurring themes in the literature—purchase prediction, browsing/comparison behavior, Indian e-commerce behavior, omnichannel behavior and review-based insights—and also highlighted a relative gap in combining those analytics with explicit DBMS/system-design mechanisms. [\[filecite\turn3file1\L132-L139\]](#)

Important academic positioning: the broad idea of AI-based customer analytics is not itself novel. The research/patent opportunity should therefore focus on a specific technical mechanism, such as event-driven behavioral scoring, problem prioritization, adaptive database optimization, privacy-aware analytics, or a validated behavior-to-intervention pipeline. The earlier research report lists several candidate directions and explicitly notes that patentability must be supported by novelty/prior-art analysis. [\[filecite\turn3file5\L299-L305\]](#)

This report gives the team a practical blueprint: project scope, use cases, architecture, technology stack, requirements, database design, AI modules, development roadmap, evaluation plan, research direction and presentation strategy.

2. Problem Statement & Objectives

Problem statement

E-commerce systems generate large volumes of customer interaction data, but raw transactions and standard dashboards often do not explain why customers stop purchasing, where conversion is lost, which segments are becoming risky, or what intervention should be tested. The proposed application addresses this by integrating behavioral data in a relational DBMS and building an explainable analytics-to-decision pipeline.

Primary objective

Develop a desktop decision-support application that analyzes e-commerce customer behavior and assists a business user in identifying sales/marketing problems, understanding contributing factors, predicting likely outcomes and selecting data-supported actions.

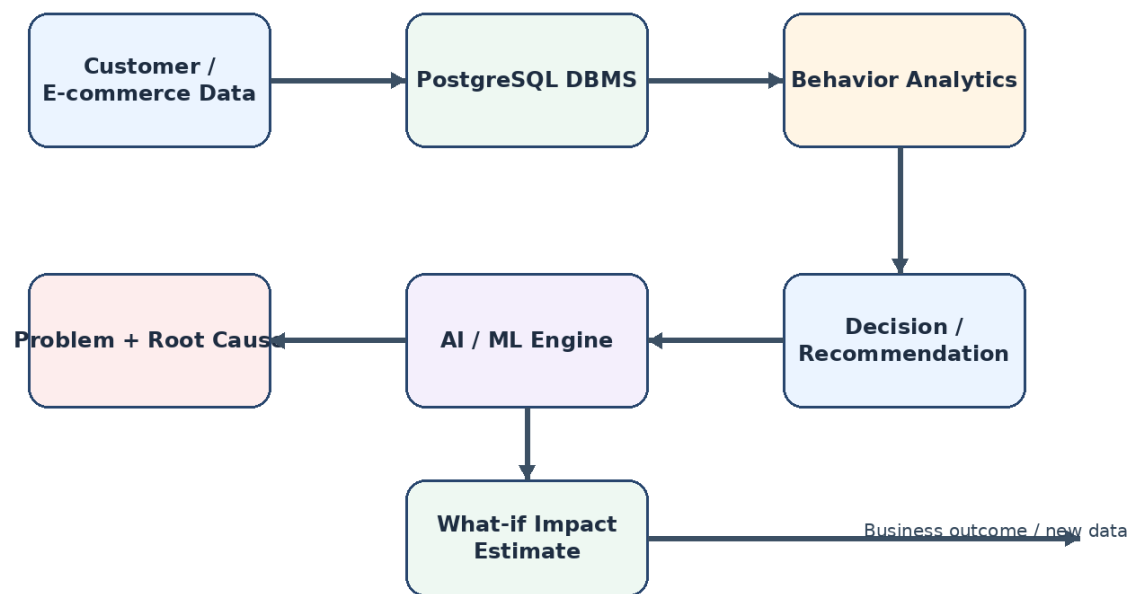
Specific objectives

- Design a normalized relational schema for customer behavior and transactions.
- Capture or simulate customer events such as search, product view, comparison, cart addition, checkout, purchase, review and return.
- Implement SQL-based analytics including RFM, conversion funnels, cohorts, product affinity and segment analysis.
- Build one or two focused ML models such as purchase, churn or cart-abandonment prediction.
- Create a problem-detection and root-cause analysis layer that combines metrics and behavioral signals.

- Generate prioritized recommendations and optional what-if impact estimates.
- Evaluate database performance and model quality against clearly defined baselines.
- Document a research gap and assess whether a specific mechanism merits patent/prior-art investigation.

3. What the Application Actually Does

Dynamic Project Flow: Data → Insight → Prediction → Action



The application should behave like a virtual business analyst. A manager can select a business area, customer segment, product category or time period. The system queries the DBMS, computes behavioral indicators, runs relevant prediction models and returns a structured explanation.

Recommended output format: Problem → Evidence → Affected segment → Likely contributing factors → Prediction → Recommended action → Estimated impact → Confidence.

This makes the system dynamic even without building a live public e-commerce website. The team can use a public dataset plus a controlled event simulator that continuously inserts new events into the database. The dashboard/application then refreshes analytics from the changing database state.

4. Real-Life Use Cases

Use case	How the system helps	Business value
Cart abandonment	Finds segments with unusually high cart-to-purchase drop-off	Higher conversions, targeted checkout investigation
Sales decline	Detects revenue/conversion changes and drills down by product/channel	Faster diagnosis of sales problem
Customer churn	Identifies customers whose recency/frequency/activity has deteriorated	Retention campaigns for high-risk customers
Product performance	Finds products with high views but weak purchases, high returns	Product prioritization and redesign investigation
Marketing effectiveness	Compares campaign exposure, engagement and purchase outcomes	Better targeting and reduced wasted marketing spend
Demand signals	Uses searches, views, wishlists and carts as early signals before purchase	Inventory and promotion planning
Review-driven issues	Links ratings/sentiment with product and sales behavior.	Detect quality/service problems earlier.

Presentation example: “The system noticed that electronics views increased while mobile checkout completion decreased. It identifies new customers as the most affected segment and recommends investigating payment friction or delivery costs before launching a broad discount campaign.”

5. How It Is Different From Ordinary Analytics Applications

Typical analytics/reporting	Proposed application
Mostly reports historical KPIs.	Detects changing behavior and business problems.
Charts answer “what happened?”.	Pipeline attempts to answer “what happened, why, what next, and what to test?”.
Separate dashboards for sales/customer data.	Unified relational behavior model links customer, session, event, cart, order, product.
Generic recommendations.	Recommendations are grounded in database evidence and model outputs.
AI is often treated as a separate component.	AI/ML is integrated with a DBMS-centered analytical workflow.
No explicit impact simulation.	Optional what-if module estimates potential impact under stated assumptions.
Innovation is usually claimed from the model alone.	Research contribution can focus on the end-to-end decision mechanism and DBMS.

6. System Architecture



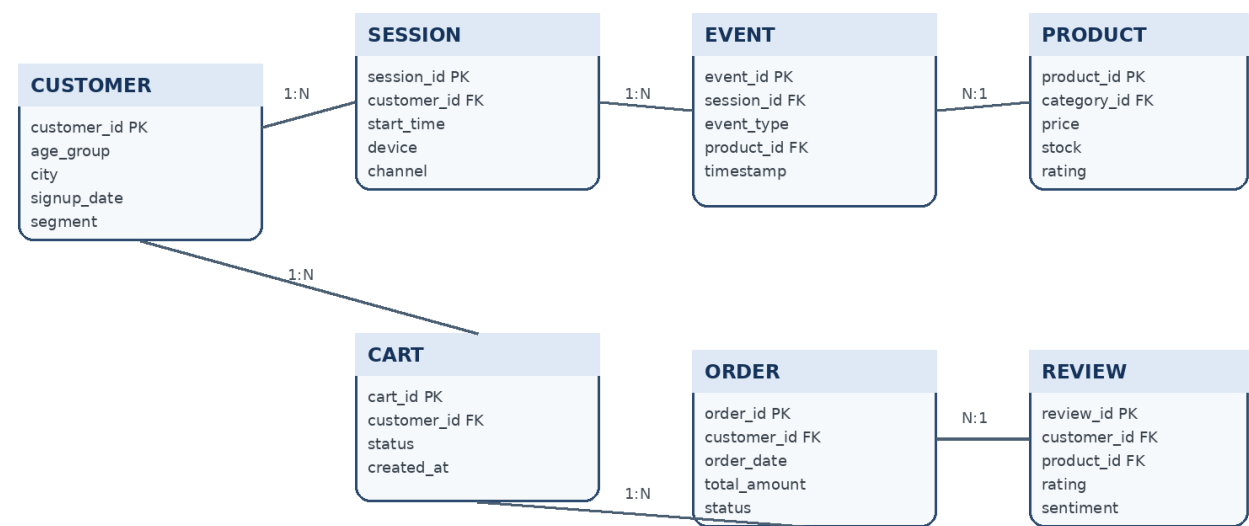
The architecture shown above is a conceptual prototype. For the actual 3rd-semester implementation, the team can simplify it to a desktop application that communicates with PostgreSQL through a Python application layer. The research report also describes a relational design containing CUSTOMER, PRODUCT, SESSION and EVENT entities, with order and order-item tables and DBMS features such as indexes, materialized views, transactions and triggers. [filecite]turn3file4[L191-L213]

Recommended logical layers

- Data layer: PostgreSQL relational database.
- Ingestion layer: CSV/data import plus optional event simulator.
- Analytics layer: SQL queries, views, materialized summaries and Python analytics.
- AI/ML layer: prediction and segmentation models.
- Decision layer: problem detection, evidence aggregation, prioritization and recommendation.
- Application layer: PySide6 desktop GUI with dashboards, alerts, predictions and reports.

7. Database Design & DBMS Requirements

Simplified Relational Schema / ER View



Core tables

Table	Purpose
CUSTOMER	Customer profile, segment and registration information.
SESSION	Shopping session, device/channel and timestamps.
EVENT	Unified behavioral events such as search, view, compare, cart and purchase.
PRODUCT	Product, category, price, rating and inventory.
CART / CART_ITEM	Cart state and products added/removed.
ORDER / ORDER_ITEM	Purchases and line-item details.
REVIEW	Ratings, review text and derived sentiment/topic fields.
PAYMENT / RETURN	Transaction and post-purchase behavior.
CAMPAIGN / CAMPAIGN_INTERACTION	Marketing exposure and response.

- Normalization: target 3NF where appropriate and explain functional dependencies.
- Constraints: primary keys, foreign keys, unique constraints, NOT NULL, CHECK constraints.
- Indexes: customer_id, product_id, timestamp, event_type and common analytical filters.
- Views/materialized views: daily sales, customer RFM, funnel summaries and segment KPIs.
- Triggers/procedures: update derived counters or behavioral scores when appropriate.
- Transactions: demonstrate ACID behavior for cart/checkout/order updates.
- Performance evaluation: compare query latency before/after selected indexes or summaries.

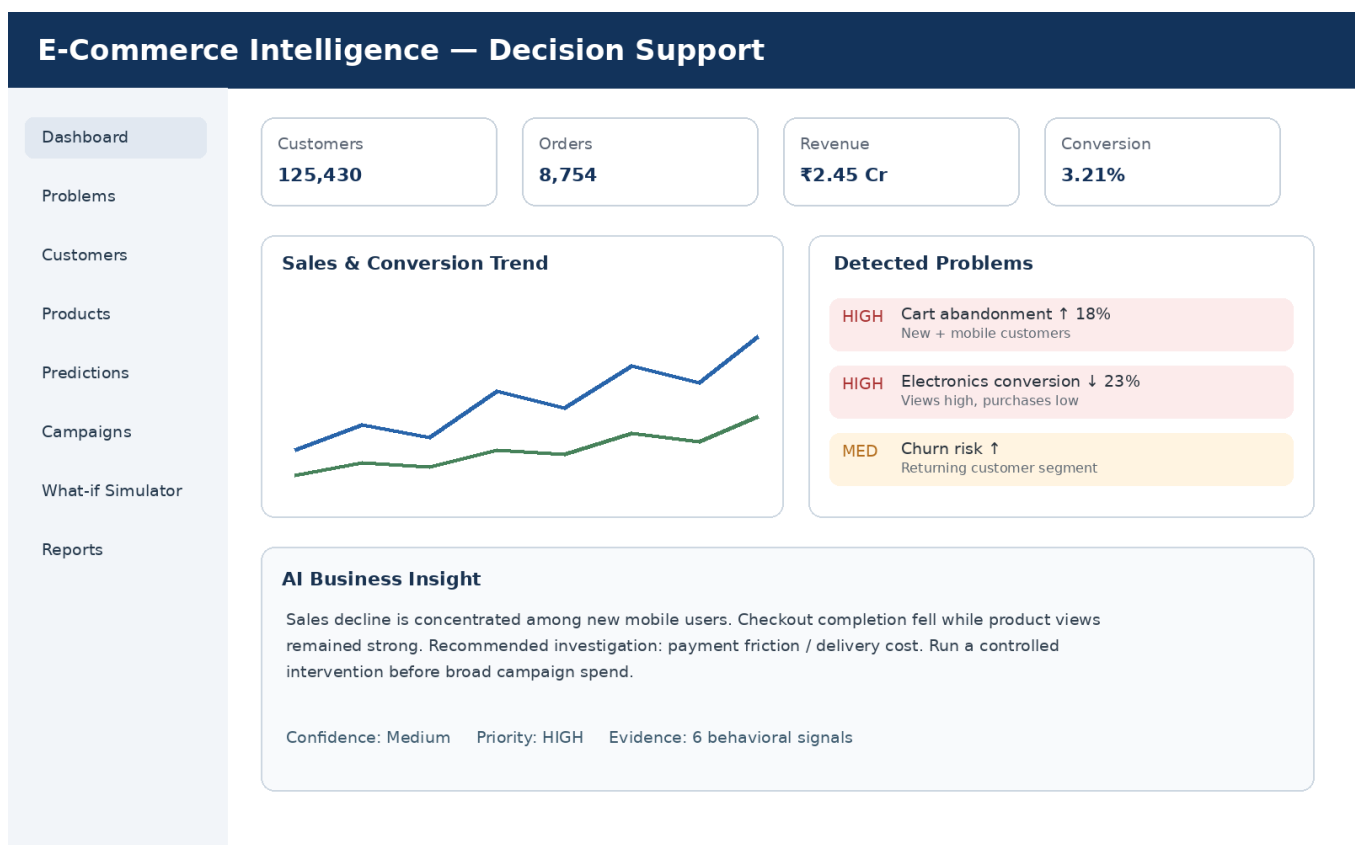
8. Technology Stack & System Requirements

Layer	Recommended technology	Reason
Database	PostgreSQL 15+	Strong relational features; ideal for DBMS demonstrations.
Language	Python 3.11+	Useful for DB connectivity, analytics and ML.

Layer	Recommended technology	Reason
GUI	PySide6	Desktop application without needing a web stack.
Analytics	Pandas, NumPy	Data cleaning and feature engineering.
ML	Scikit-learn	Classification, clustering and evaluation.
Visualization	Matplotlib / Plotly	Charts and analytical reports.
DB connectivity	SQLAlchemy / psycopg	Application-to-PostgreSQL integration.
Optional NLP	VADER / transformer model	Review sentiment or topic extraction.
Packaging	PyInstaller (optional)	Build a distributable desktop demo.

Minimum hardware: modern dual/quad-core processor, 4–8 GB RAM, 10 GB free disk space. A dedicated GPU is not required for the recommended classical ML models.

9. Application Prototype / User Experience



The desktop application can contain modules such as Dashboard, Problems, Customers, Products, Predictions, Campaigns, What-if Simulator and Reports. The UI is only a means of presenting the analytical pipeline; the academic core remains the database, analytics, algorithms and evaluation.

Recommended screen sequence for a live demonstration:

- Load or simulate new customer events.
- Open the dashboard and show KPI changes.
- Open “Problems” and select a detected issue.
- Show the SQL/analytics evidence behind the issue.
- Show the prediction model output and confidence.
- Show the recommended action and its evidence.
- Run a what-if scenario and show the estimated impact.
- Export a report containing the evidence, assumptions and recommendation.

10. How the Project Becomes Dynamic

The project does not need a public e-commerce website to be dynamic. The team can create a controlled event simulator. Each simulation step inserts events such as `product_view`, `search`, `compare`, `add_to_cart`, `checkout`, `purchase`, `review` or `return` into PostgreSQL.

As the database changes, analytical queries and derived features change. The application refreshes the affected metrics. The problem detector can then compare current behavior with historical baselines and generate an alert when a threshold or model condition is satisfied.

Example: views remain high, add-to-cart increases, but checkout completion drops. The system can flag a conversion problem, identify the affected segment/device, estimate abandonment probability, and

recommend investigating checkout/payment friction.

This approach is consistent with the research direction in the project materials, which proposes real-time/batch analytics, detailed user events and a centralized relational store.

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11. AI / Analytics Modules

Module	Input features	Output
RFM segmentation	Recency, frequency, monetary value	VIP, loyal, new, at-risk etc.
Purchase prediction	Browsing, search, cart and purchase history	Purchase probability
Cart-abandonment prediction	Cart/session/event features	Abandonment probability
Churn prediction	Recency, frequency, activity decline	Churn probability
Demand signal	Searches, views, wishlists, carts, sales	Emerging-demand score
Review analysis	Rating + review text	Sentiment/topic indicators
Problem prioritization	Severity, affected users, revenue impact, confidence	Priority score

Recommended scope: implement RFM + funnel analytics as the core; choose only two ML problems (for example purchase prediction and cart abandonment). Add churn or sentiment only if the core system is stable.

12. Problem-Solving / Decision Engine

This is the proposed differentiating layer. It should combine database evidence, analytical rules and model outputs rather than acting as a generic chatbot.

Step 1 — Detect: identify statistically or operationally unusual changes.

Step 2 — Localize: find the affected product, customer segment, channel, device or time period.

Step 3 — Explain: surface supporting behavioral signals and possible contributing factors.

Step 4 — Predict: estimate the likelihood of purchase, churn or abandonment.

Step 5 — Recommend: map the problem to candidate interventions.

Step 6 — Prioritize: rank problems/actions using severity, affected population, revenue exposure, confidence and intervention feasibility.

Step 7 — Simulate: optionally estimate the effect of a proposed intervention using historical/model evidence.

Important: recommendations should be labeled as suggestions or model estimates. They should not be presented as guaranteed sales increases or as causal proof unless a proper experiment supports the claim.

13. Development Roadmap

Recommended 12-Week Development Roadmap

1. Research & Scope	Literature review, problem definition, use cases, research gap	Weeks 1-2
2. DBMS Design	ER model, normalization, schema, constraints, indexes	Weeks 3-4
3. Data Pipeline	Dataset cleaning, ETL, event simulation, loading	Weeks 4-5
4. Analytics	RFM, funnel, cohorts, product/customer insights	Weeks 5-6
5. AI / ML	Purchase, churn or abandonment prediction; evaluation	Weeks 7-8
6. Problem Solver	Root-cause logic, prioritization, recommendation engine	Weeks 8-9
7. What-if Module	Impact estimation using historical/model evidence	Weeks 10-10
8. Application	Desktop UI, reports, integration and usability	Weeks 10-11
9. Validation & Paper	Experiments, baselines, results, documentation	Weeks 11-12

Phase 1 — Research: study purchase prediction, browsing behavior, cart abandonment, customer segmentation and review analytics. Your existing paper list includes these areas, including purchase prediction, product comparison, Indian e-commerce behavior, omnichannel behavior and review-based analysis. [filecite\turn3file0\L10-L47](#)

Phase 2 — DBMS: finalize ER diagram, schema, normalization, constraints, indexes and analytical views.

Phase 3 — Data: select a public dataset and/or create synthetic event logs. Clean and map data into the relational schema.

Phase 4 — Analytics: implement RFM, funnels, cohorts, product affinity and segment reports.

Phase 5 — AI: train and evaluate two focused models.

Phase 6 — Decision engine: implement problem detection, root-cause evidence and recommendations.

Phase 7 — Application: integrate the desktop UI, reports and what-if module.

Phase 8 — Validation: compare against baselines and document results for the paper/presentation.

14. Team Responsibilities

Member	Primary responsibility	Deliverables
Member 1 — DBMS	Schema, ER, normalization, SQL, indexes, procedures, triggers	Database + DBMS documentation
Member 2 — Application	PySide6 UI, DB connectivity, reports, module integration	Desktop application
Member 3 — Analytics/ML	RFM, funnel, feature engineering, models, evaluation	ML notebooks/scripts + metrics
Member 4 — Research/Decision Engine	Literature review, problem rules, recommendation logic, paper research	Research report + decision engine

15. Functional & Non-Functional Requirements

Functional requirements

- Import/clean e-commerce customer and transaction data.
- Store behavior events and maintain relationships through foreign keys.
- Query customers, products, orders, sessions and events.
- Calculate KPIs, funnels, RFM and customer/product segments.
- Run selected prediction models and display probability/risk outputs.
- Detect defined business problems and display evidence.
- Generate prioritized recommendations.
- Run optional what-if simulations with explicit assumptions.
- Generate/export an analytical report.

Non-functional requirements

- Correctness: referential integrity and validated calculations.
- Performance: indexed analytical queries should be benchmarked.
- Usability: a guide should be able to understand an alert within a few clicks.
- Explainability: every recommendation should show supporting evidence.
- Privacy: avoid unnecessary personally identifiable information; pseudonymize IDs.
- Reproducibility: store model version, dataset version and experiment configuration.

16. Research Paper & Patent Strategy

The project should not claim that “AI customer analytics” itself is novel. Recent research already demonstrates purchase prediction, clickstream modeling, cart-abandonment prediction and behavior-aware segmentation. For example, a 2024 Decision Support Systems paper integrates customer characteristics and browsing patterns for purchase prediction and session-level segmentation. [cite]turn0search7 A 2022 study evaluates machine-learning approaches to cart-abandonment prediction using 821,048 clickstream observations. [cite]turn0search0

A stronger research question for this project is: Does combining multi-stage customer behavior with transaction history improve business-problem detection and targeted intervention selection compared with transaction-only analytics?

Suggested experiment: compare a transaction-only baseline with a richer model containing browsing, search, cart and review signals. Report precision, recall, F1/AUC where appropriate, false-alert rate, and decision/analysis latency.

Potential technical novelty directions

- Event-driven behavioral score that updates incrementally as new customer events arrive.
- Problem-prioritization mechanism combining business impact, behavioral evidence and prediction confidence.
- Adaptive indexing/materialized-view strategy based on recurring analytical workload.
- Privacy-aware aggregate analytics that limits individual-level exposure.
- A validated behavior-to-intervention pipeline with measurable impact estimation.

The previous research report also identified unified event schemas, predictive index/cache management, privacy-preserving analytics, real-time shopping interaction scores and cross-channel fusion as candidate research/patent problem areas. [filecite]turn3file5[L299-L305] [filecite]turn3file8[L441-L447]

Patent caution: a patent cannot be assumed from an idea. Before filing, conduct a formal prior-art search and obtain guidance from your institution's IPR/patent cell or a qualified patent professional. The project should document exactly what technical mechanism is new, how it works, and how experiments demonstrate an advantage.

17. Evaluation Plan

Area	Baseline	Proposed measurement
ML prediction	Simple logistic/tree model or transaction-only features	F1s precision, recall, ROC-AUC, confusion matrix
Problem detection	Fixed generic thresholds	Detection precision, false-alert rate, time-to-detection
Database	Unindexed/basic query	Query latency before/after indexes/materialized views
Decision engine	Generic rule set	Recommendation relevance + evidence completeness
What-if module	No simulation	Calibration/error on historical or held-out scenarios

Do not fabricate results. All accuracy, latency, revenue-impact or recommendation-quality numbers in the final paper/presentation must come from your actual experiments.

18. How to Present the Project

30-second pitch: “Our application uses a relational DBMS to unify e-commerce customer behavior. It analyzes searches, browsing, cart activity, purchases and reviews, detects important business problems, predicts likely outcomes and recommends evidence-based actions. Unlike a normal dashboard, it is designed to move from data to decision.”

Live demo sequence

- Start with a clean dashboard and baseline KPIs.
- Run the event simulator or load a new time window.
- Show a KPI/behavior change.
- Open the detected problem.
- Show the affected segment and evidence.
- Show the ML prediction.
- Show the recommendation and priority.
- Run the what-if simulator.
- Finally open the SQL/query or database view used to produce the insight—this proves that the project is genuinely DBMS-centered.

Likely viva questions: Why PostgreSQL? Why normalization? Which queries are expensive? Why did you choose these features? What is the baseline? How do you avoid false alerts? How is your recommendation different from a generic AI chatbot? What is actually novel? How will you protect customer data?

19. Risks, Ethics & Limitations

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- Public datasets may not contain all event types; synthetic events should be clearly labeled as simulated.
 - Behavioral correlations do not automatically prove causation. Recommendations should be phrased as hypotheses to test.
 - Customer data should be anonymized/pseudonymized; avoid storing unnecessary names, emails or phone numbers.
 - Models can inherit dataset bias or class imbalance. Report appropriate metrics and limitations.
 - A what-if simulator is an estimate unless supported by controlled experiments.
 - Patentability is uncertain until a formal prior-art search and professional review are completed.

20. References & Research Starting Points

- [1] Rausch, T. M., Derra, N. D., & Wolf, L. (2022). Predicting online shopping cart abandonment with machine learning approaches. *International Journal of Market Research*, 64(1), 89–112. DOI: 10.1177/1470785320972526.
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- [5] An Analysis of Consumer Purchase Behavior Following Cart Addition in E-Commerce Utilizing Explainable Artificial Intelligence. *Future Internet / MDPI*, 2025.
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<https://link.springer.com/article/10.1007/s12525-024-00697-0>
- [7] Prediction of cart abandonment using imbalanced clickstream data in online shopping. *Electronic Commerce Research*, 2026.
https://ideas.repec.org/a/spr/elcore/v26y2026i4d10.1007_s10660-025-10021-3.html
- [8] Project research materials supplied in the conversation: E-Commerce Customer Behaviour Research Papers and E-Commerce Customer Behavior Analytics (DBMS Project).
Conversation-provided research material

Source note: The project plan also incorporates the team's earlier research report, which proposed a relational DBMS architecture, event-oriented behavior data, DBMS optimizations, research gaps and a 2–3 month implementation/evaluation timeline. [\[filecite\]turn3file1\[L128-L139\]](#)
[\[filecite\]turn3file6\[L320-L355\]](#)

END OF PROJECT BLUEPRINT